

## Slides 02.12: Saving and Compound Growth

Lesson 2.12, one day, 13 slides. Rates to be refreshed before teaching.

### Slide 1: Do now

- You put \$20 a week in a jar and never touch it.
- How much after one year? After 30 years?
- Show the math. Image: a clear jar with a few folded bills. Notes: \$1,040 and \$31,200. Then: same \$20 a week, into an account at 5 percent. Guess the 30-year number and write it at the top of the handout.

### Slide 2: Three reasons to save

- A goal: the SMART goal from Lesson 2.4.
- An emergency fund: a phone screen, a lost bus pass, without asking anyone.
- The future: a car, college, a first apartment, retirement. Image: three pictures in a row: a concert ticket, a cracked phone, a set of house keys.

### Slide 3: Where the money lives

- A jar or a drawer: no growth, no protection, easy to raid.
- A savings account: pays interest, insured up to \$250,000, harder to raid.
- Later, investments: more growth, real risk. Topic 2.4. Image: a jar, a bank building, a stock chart, left to right.

### Slide 4: Opening an account

- Most banks and credit unions open a youth account with an adult co-owner.
- Bring ID. Some need a small first deposit.
- Ask three things: the rate, the fees, the minimum balance.

### Slide 5: Pay yourself first

- The savings line is the first thing out of every paycheck.
- Automatic if you can set it up. Before the money hits your hand.
- Lesson 2.9 already put it second on the Budget Builder. This is why.

### Slide 6: Simple interest (yesterday)

- Pays only on the principal, the money you put in.
- \$1,000 at 5 percent: \$50 a year. Every year.
- After 3 years: \$1,150.

### Slide 7: Compound interest (today)

- Pays on the principal and on the interest already earned.
- Year 1: \$1,050. Year 2: 5 percent of \$1,050 is \$52.50, so \$1,102.50. Year 3: \$1,157.63.
- The shortcut:  $\$1,000 \times 1.05 \times 1.05 \times 1.05$ . That is 1.05 to the third power.

- Same formula as the credit card. Now it works for you.

### Slide 8: Build it by hand

- \$20 a week is \$1,040 a year. Deposit at the start of the year.
- At the end of the year the bank pays 5 percent on everything in the account.
- Five rows. Then the jar. Then the difference.
- Pairs. Calculators. 12 minutes. Notes: The check at your desk: year 2 balance before interest is \$2,132. If a pair has \$52 interest again in year 2, they took 5 percent of the new deposit alone and left out the \$1,092 carried from year 1.

### Slide 9: The reveal

- 5 years: jar \$5,200. Account \$6,033.99.
- 10 years: jar \$10,400. Account \$13,735.05.
- 30 years: jar \$31,200. Account \$72,551.18.
- How far off was your guess? Image: two bars for each row, the jar bar and the account bar, the account bar more than double at 30 years.

### Slide 10: Two savers

- Nia: \$1,000 at age 15. Never adds a dollar. At 65: about \$11,467.
- Cole: \$2,000 at age 35. At 65: about \$8,644.
- Half the money. More at the end.
- What mattered more, the amount or the years?

### Slide 11: Think about it

- A jar pays nothing but you can see it. An account pays 5 percent but it is a number on a screen. Which would you actually keep filling?
- Leo's card compounded against him at 24 percent. Your account compounds for you at 5. Which side do you want to be on?

### Slide 12: Your plan, three sentences

- How much a week.
- Where it lives, and why.
- Your first goal, and by when.

### Slide 13: Exit card

- 3 reasons to save
- 2 differences between a jar and a savings account
- 1 number: \$20 a week for 30 years at 5 percent
- Tomorrow: a trading game with candy, and a quiz. Bring everything from this topic.